

MATH 201 – Problem Set 1: Linear Algebra

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Assignment Instructions

- Show all work for full credit.
- You may collaborate, but write up solutions independently.
- Late submissions will receive a 10% penalty per day.

Problem 1: Vector Spaces

Let V be a vector space over a field F . Prove the following properties:

1. The zero vector $\mathbf{0}_V$ is unique.
2. For each $\mathbf{v} \in V$, the additive inverse $-\mathbf{v}$ is unique.
3. For any $\mathbf{v} \in V$, we have $0 \cdot \mathbf{v} = \mathbf{0}_V$ where $0 \in F$.

Solution:

Part (a): See Theorem 1 below.

Part (b): Suppose $\mathbf{w}$ and $\mathbf{w}'$ are both additive inverses of $\mathbf{v}$. Then:

$$\mathbf{w} = \mathbf{w} + \mathbf{0} = \mathbf{w} + (\mathbf{v} + \mathbf{w}') = (\mathbf{w} + \mathbf{v}) + \mathbf{w}' = \mathbf{0} + \mathbf{w}' = \mathbf{w}' \quad (1)$$

Part (c): We have $0 \cdot \mathbf{v} + 0 \cdot \mathbf{v} = (0 + 0) \cdot \mathbf{v} = 0 \cdot \mathbf{v}$. Adding $-(0 \cdot \mathbf{v})$ to both sides gives $0 \cdot \mathbf{v} = \mathbf{0}_V$.

Definition 1 *Vector Space.*

A **vector space** over a field F is a set V together with two operations:

- **Addition:** $+: V \times V \rightarrow V$
- **Scalar multiplication:** $\cdot: F \times V \rightarrow V$

satisfying the eight vector space axioms.

Theorem 1 *Uniqueness of Zero.*

In any vector space V , the zero vector $\mathbf{0}_V$ is unique.

Proof. Suppose there exist two zero vectors $\mathbf{0}$ and $\mathbf{0}'$ in V .

By the definition of zero vector:

$$\mathbf{0} + \mathbf{0}' = \mathbf{0}' \quad (\text{since } \mathbf{0} \text{ is a zero vector}) \quad (2)$$

$$\mathbf{0} + \mathbf{0}' = \mathbf{0} \quad (\text{since } \mathbf{0}' \text{ is a zero vector}) \quad (3)$$

Therefore $\mathbf{0} = \mathbf{0}'$, proving uniqueness. ■

Problem 2: Comparison of Vector Spaces

Compare and contrast the following vector spaces.

Definition *Real Vector Space*.

A vector space over $\mathbb{R}$.

Example: $\mathbb{R}^n$ with standard operations.

Properties:

- Ordered field
- Complete (has limits)
- Infinite dimensional examples exist

Definition *Complex Vector Space*.

A vector space over $\mathbb{C}$.

Example: $\mathbb{C}^n$ with standard operations.

Properties:

- Algebraically closed
- Every polynomial factors
- Important in quantum mechanics

Problem 3: Linear Independence

Prove properties related to linear independence.

Lemma 1 *Steinitz Exchange*.

If S is a linearly independent set and T is a spanning set with $|S| > |T|$, then S cannot be linearly independent. Hence $|S| \leq |T|$.

Proof. The proof proceeds by induction on $|S|$. The base case is trivial. For the inductive step, we exchange elements between S and T while preserving the spanning property... □

Corollary 1 *Dimension is Well-Defined*.

All bases of a finite-dimensional vector space have the same cardinality.

Example 1 *Basis of $\mathbb{R}^3$* .

The standard basis of $\mathbb{R}^3$ is:

$$\mathcal{B} = \{e_1, e_2, e_3\} = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \quad (4)$$

This basis has exactly 3 elements, so $\dim(\mathbb{R}^3) = 3$.

Problem 4: Matrix Properties

Consider the following matrices and their properties.

Solution:

The rank-nullity theorem states: $\text{rank}(A) + \text{nullity}(A) = n$ for an $n \times n$ matrix.

For matrix A : $2 + 0 = 2$ ✓

For matrix B : $1 + 1 = 2$ ✓ (singular matrix)

For I_2 : $2 + 0 = 2$ ✓

Matrix	Rank	Nullity	Determinant
$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	2	0	-2
$B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$	1	1	0
$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	2	0	1

Properties of 2×2 matrices

Problem 5: Computational Linear Algebra

Implement Gaussian elimination in Python.

Solution:

```

1 import numpy as np
2
3 def gaussian_elimination(A, b):
4     """Solve Ax = b using Gaussian elimination."""
5     n = len(b)
6     # Augmented matrix
7     Ab = np.hstack([A.astype(float), b.reshape(-1, 1)])
8
9     # Forward elimination
10    for i in range(n):
11        # Pivot
12        max_row = np.argmax(np.abs(Ab[i:, i])) + i
13        Ab[[i, max_row]] = Ab[[max_row, i]]
14
15        for j in range(i + 1, n):
16            factor = Ab[j, i] / Ab[i, i]
17            Ab[j, i:] -= factor * Ab[i, i:]
18
19    # Back substitution
20    x = np.zeros(n)
21    for i in range(n - 1, -1, -1):
22        x[i] = (Ab[i, -1] - np.dot(Ab[i, i+1:n], x[i+1:])) / Ab[i, i]
```

```

23
24     return x

```

Problem 6: Eigenvalues

Find the eigenvalues of $A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$.

Solution:

$$\det(A - \lambda I) = \det \begin{pmatrix} 3 - \lambda & 1 \\ 0 & 2 - \lambda \end{pmatrix} = (3 - \lambda)(2 - \lambda) = 0 \quad (5)$$

Therefore $\lambda_1 = 3$ and $\lambda_2 = 2$.

Theorem 2 *Characteristic Polynomial.*

The eigenvalues of A are the roots of $\det(A - \lambda I) = 0$.

Problem 7: Proof Styles

Demonstrate different QED symbol styles.

Proof. This proof uses the filled square (default).



Proof. This proof uses the hollow square.



Proof. This proof uses the filled cube.



Proof. This proof uses the classical Q.E.D.

Q.E.D.